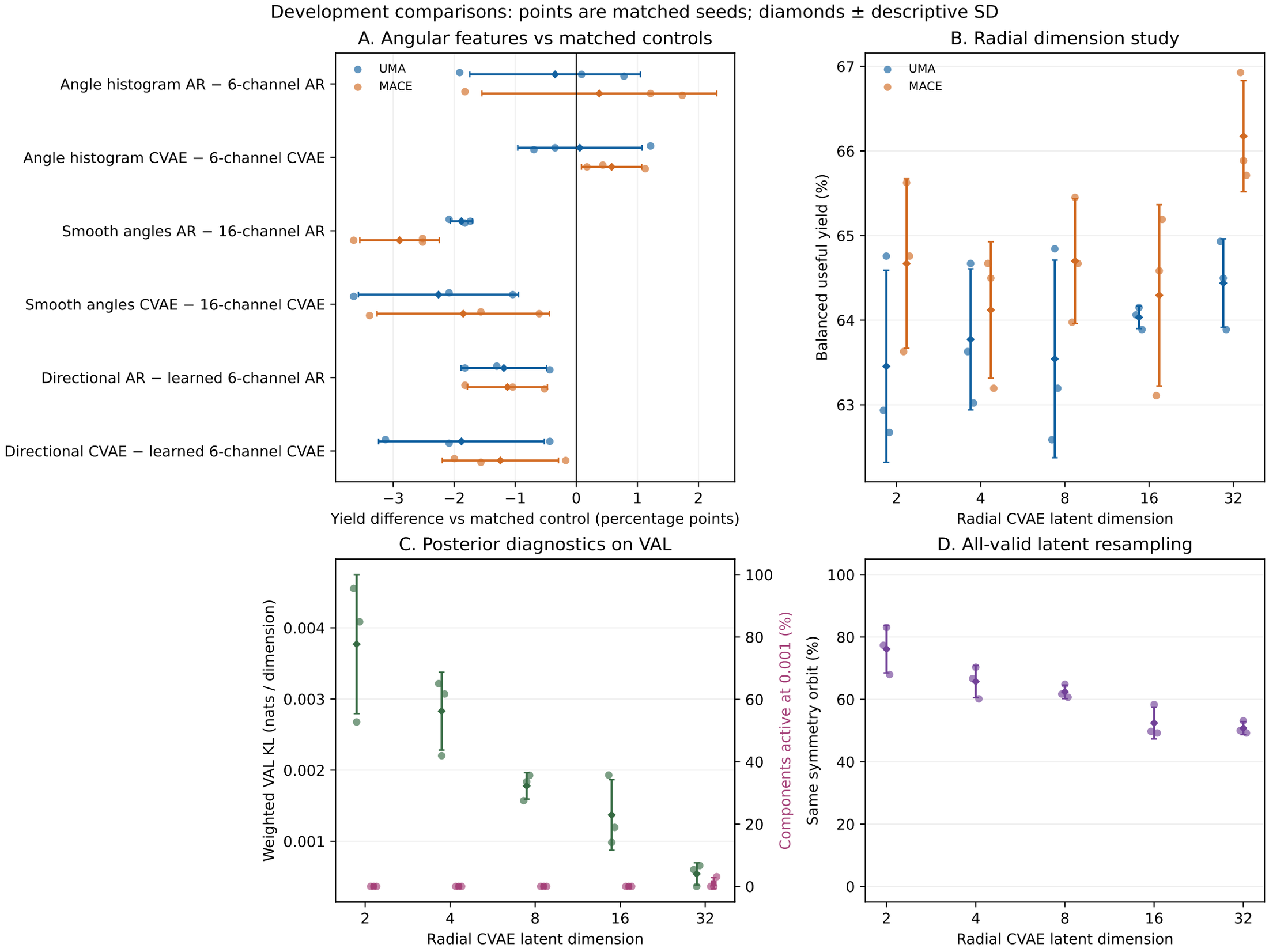
Development comparisons: points are matched seeds; diamonds ± descriptive SD
A. Angular features vs matched controls
UMA
MACE
Angle histogram AR − 6-channel AR
Angle histogram CVAE − 6-channel CVAE
Smooth angles AR − 16-channel AR
Smooth angles CVAE − 16-channel CVAE
Directional AR − learned 6-channel AR
Directional CVAE − learned 6-channel CVAE
−3
−2
−1
0
1
2
Yield difference vs matched control (percentage points)
B. Radial dimension study
UMA
MACE
Balanced useful yield (%)
67
66
65
64
63
2
4
8
16
32
Radial CVAE latent dimension
C. Posterior diagnostics on VAL
Weighted VAL KL (nats / dimension)
0.004
0.003
0.002
0.001
Components active at 0.001 (%)
100
80
60
40
20
0
2
4
8
16
32
Radial CVAE latent dimension
D. All-valid latent resampling
Same symmetry orbit (%)
100
80
60
40
20
0
2
4
8
16
32
Radial CVAE latent dimension